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Inter-satellite link technologies and applications in Low Earth Orbit satellite networks

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ABSTRACT

To support the objectives of next-generation networks, Low Earth Orbit (LEO) satellite communications have attracted significant attention. In large-scale LEO constellations, Inter-Satellite Links (ISLs) are crucial for satisfying a range of network requirements. This work provides a comprehensive overview of ISLs from three perspectives. First, we introduce the ISL communication technologies, with an emphasis on satellite constellations and laser-based ISLs. Next, we explore the role of ISLs in traffic delivery within satellite networks, focusing on topology control, traffic scheduling, and routing strategies. Finally, we highlight key applications of ISL-enabled network systems, including their integration with Global Navigation Satellite Systems (GNSS).

1. Introduction

Inter-Satellite Link (ISL) is a fundamental technology for modern LEO satellite communication systems. LEO mega-constellations, such as Starlink, offer notable advantages in terms of low latency and cost efficiency [1], and are already delivering global Internet services. Within these systems, ISL networking plays a pivotal role in enhancing network performance and operational efficiency. However, ISLs in satellite networks face distinct challenges arising from the unique conditions of the space environment. In this work, we examine the state-of-the-art ISL technologies and explore their applications and roles within LEO satellite networks.

Communication satellites continuously orbit the Earth, and it is essential to account for the dynamic nature of ISL topology and resource allocation. Traditionally, Radio Frequency (RF)-based ISLs have been used, offering broad coverage with relatively few satellites. However, this approach comes with limitations, such as restricted bandwidth and vulnerability to interference. To address these challenges, Laser ISLs (LISLs) have emerged as a promising alternative, offering higher bandwidth and stronger immunity to interference. However, LISLs demand greater power and precise optical alignment over long distances, making their implementation more technically challenging. In LEO constellations, the shorter inter-satellite distances significantly reduce transmission power requirements and simplify the Pointing, Acquisition, and Tracking (PAT) process. As a result, LISLs are particularly well-suited for large-scale LEO constellations and are now the preferred choice compared to Geostationary (GSO) or Medium Earth Orbit (MEO) systems.

In addition to the technical challenges of establishing LISLs, the rapid orbital motion of LEO satellites further complicates network design. Specifically, relying on a static ISL topology becomes increasingly impractical, as it cannot effectively adapt to frequent and dynamic changes in satellite positions and available resources. Therefore, ISL networking in LEO systems must be designed with flexibility and adaptability in mind, responding to the evolving topological and resource constraints. A variety of solutions have been proposed to address the challenge. However, most of these works primarily focus on free-space optical communications, general ISL technologies, or LEO mega-constellation architectures [2–5], and the networking aspects of ISLs are underexplored. In this survey, we aim to fill this gap by focusing on ISL networking in LEO satellite systems. We explore the key elements of inter-satellite communication, including transmission technologies (e.g., RF and Laser), resource limitations, and the implications of dynamic topology changes in LEO environments.

The structure of this paper is organized around the following three key thrusts, as illustrated in Fig. 1.

- **ISL Communications:** We provide an overview of ISL communication technologies, with an emphasis on emerging Laser ISL (LISL) advancements. This includes a discussion of the operating environment, various satellite orbits, and constellation architectures relevant to ISL deployment.

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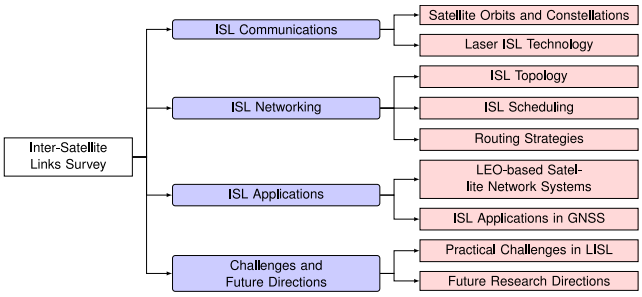


Fig. 1. Structure of the paper.

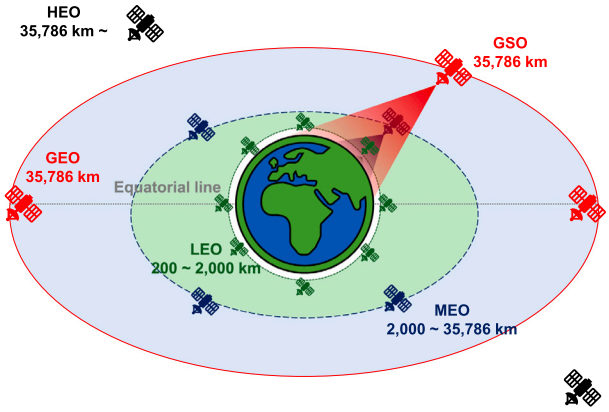


Fig. 2. Classification of satellite orbits.

- **ISL Networking:** We examine the end-to-end processes involved in ISL-based networking, covering key aspects such as topology design, link scheduling, and routing strategies. Special attention is given to recent and active research directions in this area.
- **Applications of ISL:** We explore the practical applications of ISLs in enhancing overall satellite network performance, with a particular focus on their integration into modern Global Navigation Satellite Systems (GNSS) and related infrastructures.

Finally, we synthesize the practical challenges and outline future research directions to provide a roadmap for next-generation satellite networks. Through these three focal areas, this paper aims to provide a comprehensive understanding of ISL technologies and their networking potential, with the goal of guiding future research and development in LEO satellite communications.

2. ISL communications

We introduce the basic foundation of ISL communications. Starting from satellite orbits and constellations, we provide a brief overview of laser ISL technology that has the advantages of interference-free communications and wide bandwidth compared to conventional RF-based ISL.

2.1. Satellite orbits and constellations

To understand inter-satellite communications, it is essential to consider the orbital environments in which satellites operate. The characteristics of satellite orbits directly influence communication performance, link feasibility, and network design strategies. In this section, we begin by introducing the different types of satellite orbits and constellation structures that form the foundation for ISL deployment.

Satellites orbit the Earth along a geocentric trajectory, which is typically classified into four types based on their altitude: Low Earth Orbit (LEO), Medium Earth Orbit (MEO), Geosynchronous Orbit (GSO) and Geostationary Earth Orbit (GEO), and High Earth Orbit (HEO). Fig. 2 illustrates satellite orbits and coverages by altitude. These regimes trade off coverage, latency, and complexity, impacting ISL design. Accurate orbit geometry and prediction analysis is key to understanding these trade-offs.

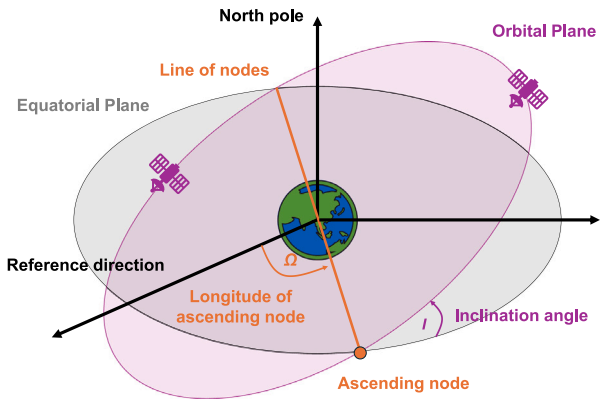


Fig. 3. Orbital plane.

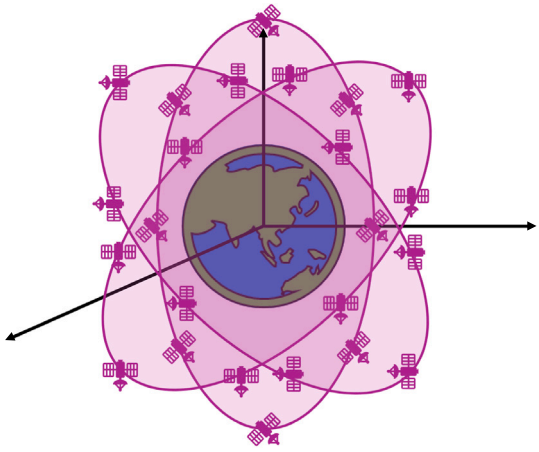


Fig. 4. Walker delta constellation ($N/P/F$: 24/3/0).

Each satellite orbits within an Orbital Plane (OP), a two-dimensional plane through Earth's center, defined by inclination angle (I) and ascending node longitude (Ω). Fig. 3 depicts OP geometrically as spherical angular coordinates. Orbit propagation predicts positions for ISL connectivity management, enabling availability estimation and link scheduling; recently, machine learning-based methods enhance accuracy and reduce overhead [6–8].

To achieve effective coverage and continuous ISL connectivity, satellites deploy in constellations across multiple orbital planes, organized into shells of shared altitude and inclination. The Walker constellation [9] is widely used, defined by total satellites N, orbital planes P, and phase offset F, ensuring consistent altitude and inclination for efficient deployment.

Walker constellations are classified into delta ($i \approx 60°$) and star ($i \approx 90°$) configurations, with delta lacking polar coverage. The Walker delta (e.g., $N = 24$, $P = 3$, $F = 0$) is most common due to global coverage and simplicity, as shown in Fig. 4. Simulation-based studies compare constellation schemes [10], focusing on performance-cost balance, coverage maximization, ISL range/delay minimization, or deployment cost reduction. Zhang et al. [11] evaluate single-shell LEO constellations, revealing latency degradation below 28 satellites per orbit and minimal end-to-end delay when orbital inclination aligns with

Table 1
Types and characteristics of orbital-plane LISLs.

Type	Relative motion	Link duration	Network role
Intra-OP	Nearly static	Permanent	Backbone
AOP/NOP	Stable pattern	Quasi-permanent	Mesh
COP	High velocity	Temporary	Shortcut

endpoint geographic angles, resulting in more intra-orbit and fewer zig-zag inter-orbit hops.

To effectively study ISL networking in satellite constellations, it is essential to consider the dynamic topology caused by the continuous movement of satellites in their orbits. Accurate simulation requires not only modeling of satellite motion but also a 3D visualization of the Earth and satellite positions to properly evaluate the networking performance of the system. To support such analysis, Hypatia has been introduced as a platform for simulating and visualizing ISLs in LEO mega-constellations. It enables realistic modeling of satellite trajectories and inter-satellite connectivity over time, providing valuable insights into the design and optimization of dynamic satellite networks [12].

2.2. LISL technology

Building on the dynamic nature of satellite constellations and ISL networking, the choice of transmission technology becomes critical. ISL transmissions can generally be categorized into two types: Radio Frequency (RF) and Free-Space Optical (FSO) transmissions. The latter is commonly referred to as Laser ISL (LISL), which is implemented through Optical Wireless Communication (OWC). Since RF ISLs face significant limitations, such as licensed spectrum requirements, high susceptibility to interference, and lower energy efficiency [1], we pay our attention to LISLs, despite their longer link setup time.

LISLs can be broadly classified into intra-orbital-plane (intra-OP) and inter-orbital-plane (inter-OP) types, depending on their relative positioning of the satellites. In intra-OP LISLs, satellites share the same altitude and orbital plane, moving at nearly identical velocities. This results in relatively small positional changes between the satellites, making the alignment and maintenance of the laser link more stable and consistent. Consequently, intra-OP LISLs are typically maintained over longer durations and are referred to as permanent LISLs. On the other hand, inter-OP LISLs connect satellites in different orbital planes, where differences in velocity and relative position introduce additional challenges. These inter-OP LISLs can be further characterized based on spatial relationships as Adjacent Orbital Plane (AOP), Nearby Orbital Plane (NOP), and Crossing Orbital Plane (COP) links. Due to the higher degree of relative motion, most inter-OP LISLs are considered temporary. However, in some cases, AOP and NOP LISLs can exhibit stable geometric patterns over time and may also be treated as quasi-permanent links in practice [13]. Table 1 summarizes the types and characteristics of orbital-plane LISLs. Understanding these classifications is essential for designing efficient link scheduling and routing strategies, especially in dynamic LEO constellations where maintaining connectivity under changing topologies is critical.

Topologically, intra- and inter-OP LISLs play complementary roles: intra-OP LISLs sustain local optical meshes, while inter-OP LISLs provide crucial shortcuts that significantly reduce end-to-end latency. Simulation results based on Starlink Phase I parameters (550 km altitude, 53° inclination) quantify this distinction [13,14]. Intra-OP LISLs maintain a constant range and fixed free-space path loss, ensuring stable link budgets for permanent connectivity [14]; theoretically, a single satellite could establish up to 88 such permanent links if the range extends to 5016 km. Conversely, inter-OP LISLs — particularly those between crossing planes — are established for short intervals (e.g., approximately 283 s) [13] and exhibit time-varying path loss with fluctuating Signal-to-Noise Ratios due to rapid relative motion [14]. Despite these stability challenges, integrating these temporary links decreases the average path length, thereby reducing inter-continental latency by on the order of 10–20 ms [15]. In terms of feasibility and capacity, practical implementations have been demonstrated with the European Data Relay Satellite System (EDRS) achieving 1.8 Gbps links between GEO and LEO, while experimental LEO-LEO homodyne BPSK transmissions have reached 5.6 Gbps [3]. Furthermore, while current terminals support 10 Gbps over 4000 km, advanced techniques combining wavelength division multiplexing (WDM) with orbital angular momentum (OAM) show potential for aggregate capacities reaching 100 Tbps [16].

2.2.1. Beam management

Establishing a LISL requires that each satellite be equipped with a Laser Communication Terminal (LCT), which includes a laser transceiver capable of sending and receiving optical signals. The performance and reliability of LISLs are heavily influenced by how effectively the LCT manages its optical beam. To maximize transmission efficiency, advanced beam management techniques tailored for dynamic satellite environments are essential.

Beam management for LISL typically encompasses three core functions: beamforming, beam switching, and beam steering. Beamforming enhances the strength and focus of the optical signal by adjusting the antenna array configuration to optimize directional transmission. Beam switching enables the LCT to redirect or reassign the beam's direction or frequency to establish new ISLs. In contrast, beam steering plays a critical role in maintaining alignment with a target satellite by continuously adjusting the beam's orientation in response to the satellite's relative motion.

Recent studies have investigated the performance of beam switching and electronic beam steering for inter-OP LISLs. The findings suggest that beam steering offers superior performance by increasing the average transmission speed, making it a promising approach for future LISL deployments in dynamic satellite constellations [17].

2.2.2. Pointing, acquisition, and tracking

Building on the importance of beam management, establishing a new LISL involves a critical three-stage process known as Pointing, Acquisition, and Tracking (PAT). This procedure is essential for initiating and maintaining stable optical communication between satellites [18]. Fig. 5 shows the PAT process in two orbital planes. Pointing refers to orienting the laser beam toward the intended target satellite, while acquisition involves detecting and locking onto the incoming optical signal from that satellite. Once communication is initiated, tracking continuously adjusts the beam alignment to compensate for relative satellite motion, ensuring a stable connection throughout the data exchange process [19].

In dynamic LEO constellations, the PAT process introduces non-negligible latency. While mechanical pointing can induce substantial delays due to large slew angles, the acquisition phase itself is also a critical bottleneck. Validated models utilizing data from NASA missions, such as TBIRD, indicate that this acquisition process typically requires between 30 and 60 s to successfully lock onto the signal [20]. However, this duration varies significantly depending on the orbit type and the specific phase of the setup process. For instance, experimental data shows that acquisition times can range from 8 s for initial spatial acquisition in LEO-LEO missions to 55 s for scanning the uncertainty cone in GEO-LEO links [21]. Furthermore, mechanical factors such as satellite vibrations from thrusters and reaction wheels can adversely affect the PAT procedure. These disturbances extend the time required to scan and acquire the target beam, thereby narrowing the effective communication window and necessitating adaptive beam divergence control [22].

Integrating these acquisition times into routing and scheduling is crucial because PAT delays substantially reduce effective contact duration and alter edge costs. If ignored, these delays lead to inefficient path selection, underestimated end-to-end latency, and overestimated

network capacities [20]. Notably, the TerraSAR-X–NFIRE experiments demonstrated that the full communication process — including spatial acquisition and phase locking — typically required 41 to 52 s to reach steady operation [23]. Such delays necessitate substantial buffering capacity to prevent data loss during topology transitions [21]. Consequently, long establishment times can induce route convergence delays and temporary isolation, potentially causing service interruptions on the order of tens of seconds [23]. To mitigate these issues, heuristic on-demand routing algorithms have been developed that minimize the average end-to-end latency, where the LISL setup delay is modeled as a penalty term in the latency cost, thus implicitly trading off shorter paths against the cost of establishing new dynamic links [24]. While Wang et al. [16] note that temporary LISLs in large LEO constellations can enhance routing flexibility and reduce node delay, the associated switching complexity and setup delays imply that their use must be carefully scheduled to balance connectivity needs with power efficiency. Moreover, fully on-demand sleep/activation control of LISLs is often impractical: the LISL setup (pointing) delay, typically on the order of seconds, is intolerable for traffic with strict latency requirements, and such aperiodic state changes may compromise routing reliability [25].

Operational overhead and latency vary significantly depending on the dynamic establishment scenario, specifically distinguishing between reconnecting to a previous link and switching to a new target. Reconnecting to the same satellite pair typically incurs lower latency because terminals can leverage prior orbital parameters to restrict the search space within the Field of Uncertainty (FOU) without large angular slewing [20]. For instance, commercial terminals such as Mynaric's CONDOR can reduce acquisition time from an initial 30 s to approximately 2 s for subsequent setups [26]. However, maintaining these connections requires continuous compensation for tracking errors and distance changes to avoid unexpected disruptions caused by misalignment or vibrations [27]. Since LISLs often have only short available link durations and can be unexpectedly disrupted by pointing errors or satellite vibrations, minimizing acquisition time under such disturbances is critical for rapid link re-establishment and robust link maintenance [22]. To alleviate the immediate packet loss caused by link failures, the Disruption Tolerant Distributed Routing (DTDR) algorithm utilizes local topology information to instantly calculate backup paths [28]. Furthermore, a link re-planning algorithm has been proposed to dynamically adjust the topology, aiming to maximize system capacity and restore network connectivity with minimal switching costs [29]. As the LEO satellite network with ISLs becomes more complex, especially under stochastic link failures, optimization methods such as improved Genetic Algorithms are employed to jointly optimize task revenue, link switching times, and routing cost [30]. Additionally, service-aware temporal betweenness metrics can be employed to identify critical nodes and quantitatively evaluate how satellite node failures affect service connectivity and path delays over time [31].

In contrast, switching the link target to a new satellite introduces significantly higher overhead, primarily driven by the mechanical slewing time required to align the optical head toward a new trajectory. Although this process utilizes ephemeris data to compute pointing vectors, the physical realignment results in a distribution of pointing delays that peaks at approximately 80 s in LEO-to-LEO retargeting scenarios [20]. This stands in sharp contrast to radio frequency links, which typically establish connections within 200 ms; conversely, LISLs face prohibitive delays ranging from 10 to 60 s due to the complex PAT process [32]. Experimental validation further indicates that, even with optimized scan patterns such as Spiral or Rose, the Mean Acquisition Time (MAT) for establishing a new LISL remains between 11.4 and 14.6 seconds [33]. The establishment of temporary links is particularly challenging between satellites in crossing orbital planes, where high relative velocities and repeated setup overheads complicate the process [15]. To alleviate these issues, static topology construction based on optimal phasing factors has been proposed, aiming to maximize link duration and minimize the frequency of dynamic switching [34].

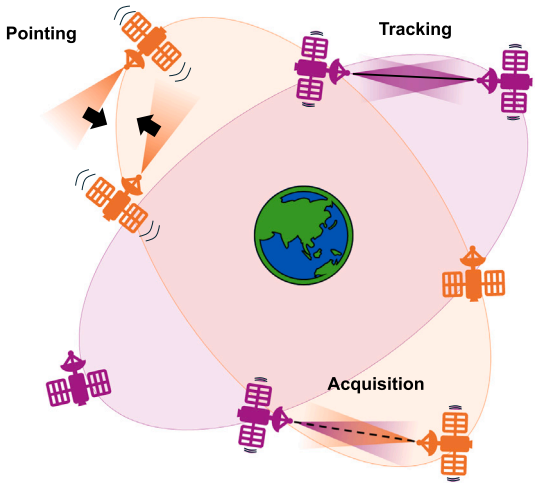


Fig. 5. Pointing, acquisition, and tracking process for LISL.

2.2.3. Signal processing

Once a LISL is successfully established, signal processing becomes a crucial aspect of ensuring reliable and efficient communication between satellites. Signal processing in ISL systems can be broadly divided into three key components: channel coding, multiple access, and modulation. The choice of technology in each category must be carefully tailored to the power constraints and physical limitations of satellite platforms [35].

In laser communication systems, modulation and detection techniques critically determine transmission efficiency and signal robustness. Primary approaches include Intensity Modulation with Direct Detection (IM/DD), which modulates light intensity for direct detection without phase coherence, and coherent systems, which modulate phase and amplitude for superior sensitivity.

Coherent detection excels in LISL configurations over tens to hundreds of kilometers, effectively extracting attenuated signals. IM/DD commonly employs On-Off Keying (OOK) for binary pulses or Pulse Position Modulation (PPM) for temporal encoding, while coherent schemes use Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK) for high speeds, and Quadrature Amplitude Modulation (QAM) for spectral efficiency via dedicated Mach–Zehnder modulators [19]. Performance evaluations in LISL leverage Bit Error Rate (BER) and Q-factor metrics [36–39], with Wavelength Division Multiplexing (WDM) enhancing throughput [40].

In summary, ISL communication technologies have evolved to address the stringent demands of next-generation satellite networks, including high data rates, broad bandwidth, and global coverage. These goals are increasingly being realized through the adoption of laser-based ISLs within LEO mega-constellations. However, to fully harness their potential, it is equally essential to develop and integrate optimized networking strategies that can operate effectively within the dynamic and resource-constrained environment of inter-satellite communication.

3. ISL networking

In this section, we review recent research on ISL networking, with a focus on optimizing network performance under the constraints of limited onboard resources and the dynamic positional changes of satellites in orbit. Key topics include topology design, where visibility analysis and ISL planning are explored within the context of LEO mega-constellations. In addition, we examine ISL scheduling techniques, many of which employ optimization-based approaches to manage link activation and resource allocation efficiently. Finally, we introduce a

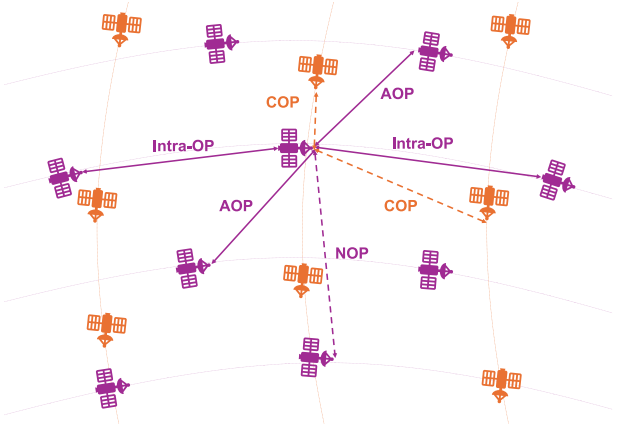


Fig. 6. Example of ISL Topology. The dotted lines represent orbits. The solid and dashed arrows represent persistent links and temporary links, respectively.

range of routing strategies that leverage ISLs, including methods based on hop count analysis as well as adaptive routing algorithms designed to cope with the continuously evolving satellite topology.

3.1. ISL topology

The ISL topology defines the network architecture established through interconnections among satellites within a constellation. Unlike terrestrial networks with fixed nodes, ISL topologies must account for the continuous orbital motion of satellites. The design of such topologies involves addressing critical system-level trade-offs, including feeder link congestion arising from traffic aggregation and latency introduced by multi-hop routing [41]. To ensure reliable connectivity and optimal network performance, the ISL topology must adapt dynamically to variations in satellite positions and real-time network conditions.

For constellations composed of MEO or GEO satellites, topology design is relatively straightforward due to the limited number of satellites and their slow orbital motion. In contrast, LEO mega-constellations presents substantial challenges for ISL topology design. These systems typically consist of thousands of satellites distributed across multiple orbital planes and shells, each traveling at velocities exceeding 27,000 km/h. The resulting high mobility and spatial density necessitate more sophisticated design strategies to ensure consistent link availability and network stability under dynamic condition [42].

3.1.1. Structural topology design

To cope with this spatiotemporal complexity, topology design is often simplified by assuming quasi-static satellite positions over short time intervals, allowing the problem to be formulated as a link assignment task. A representative method following this approach is the Finite State Automata (FSA)-based link assignment and routing technique [43], which enables systematic link configuration under predictable orbital dynamics.

Among existing ISL configurations, the grid-mesh topology is widely adopted, particularly in LEO mega-constellations. In a typical full grid-mesh structure, each satellite establishes ISLs with its nearest neighbors within the same orbital plane (intra-OP) and with adjacent orbital planes (AOPs) [44]. Despite its simplicity and ease of implementation, this topology presents inherent scalability and efficiency limitations. In standard implementations — used both in industry and academia — each satellite maintains four persistent ISLs (two intra-OP and two AOP) as shown in Fig. 6. While such persistent connectivity simplifies routing and link management, it can also result in resource underutilization and increased power consumption, particularly in highly dynamic orbit and traffic conditions.

To overcome the inherent limitations of rigid topologies, Bhattacherjee et al. [42] proposed a motif-based design methodology that generalizes connectivity patterns to optimize network efficiency and latency. Aligning with this pursuit of efficiency, recent research has further investigated structural variations of the standard grid-mesh, often termed +Grid topologies. Specifically, Schroder et al. [45] evaluated three configurations — Sparse, +Grid, and Extended +Grid — within a LEO network. Their analysis reveals that the Extended +Grid topology reduces the hop count by 50% compared to the standard +Grid while maintaining identical hardware requirements. This suggests that topological optimization can yield significant performance gains without incurring additional hardware costs.

Complementing these performance-driven designs, structural simplification has been investigated to lower hardware costs. Chen et al. [46] analyzed the routing performance of a 3-ISL topology—a configuration adopted by Starlink for its cost-efficiency. They demonstrated that this reduced topology can achieve performance comparable to the traditional 4-ISL architecture under specific constellation parameters, thereby presenting a viable solution for minimizing the number of required laser terminals.

These approaches dynamically reconfigure link patterns according to traffic demand and orbital dynamics, enhancing link utilization, network adaptability, and energy efficiency. To further address the engineering constraints of inter-plane laser pointing, more sophisticated heterogeneous architectures have been proposed. For instance, Zhang et al. [32] introduced a hybrid ISL framework that combines stable intra-orbit laser backbones with agile inter-orbit radio links. By employing a Topological Structure Unit (TSU)-based optimization strategy, this approach effectively capitalizes on the complementary strengths of each medium, validating significant reductions in end-to-end delay and hop counts across mega-constellations.

3.1.2. Visibility analysis

In LEO mega-constellations, the presence of numerous temporary ISLs and complex geometric relationships between satellites and the Earth result in a highly dynamic and unstable ISL topology. To address this, visibility analysis plays a critical role in determining which satellite pairs are capable of maintaining line-of-sight connections, taking into account both geometric constraints (e.g., orbital positioning) and physical limitations (e.g., terminal orientation, beam range). This approach enables the construction of simpler, more stable, and efficient ISL topologies. Recent studies have introduced connectivity analysis based on time-varying visibility graphs in LEO mega-constellations. For instance, a time-varying visibility matrix was proposed to model satellite-to-satellite connectivity over time, resulting in improvements in end-to-end link distances and overall path efficiency [47]. Further research has explored topology optimization using visibility analysis to derive static yet optimized topologies for LISL-based mega-constellations [34]. In addition, topological design techniques that aim to minimize ISL switching overhead through visibility-aware strategies have been proposed [48], where inter-plane connectivity, particularly relevant to large-scale constellation designs like the Walker star configuration, has also been investigated. These studies focus on understanding how satellites in different orbital planes can maintain stable ISLs over time, thereby improving topological resilience and routing flexibility.

3.1.3. ISL planning

Building upon the underlying ISL topology, ISL planning focuses on establishing and maintaining active links between satellites in response to the network's dynamically evolving structure. The goal is to ensure high throughput, efficient link utilization, and robust connectivity, despite the continuous changes in satellite positions. A range of planning algorithms have been proposed to optimize ISL assignments in this context. To minimize link switching frequency while maximizing data throughput, a deep reinforcement learning-based approach using

Deep Deterministic Policy Gradient (DDPG) was introduced, featuring centralized training and distributed execution for inter-plane ISL planning [49]. Similarly, Deep-ISL [50] leverages a parameterized Deep Q-Network (DQN) to jointly optimize inter-plane ISL configuration and per-link transmit power allocation, balancing performance and resource constraints. To reduce unnecessary link overhead, the Asymmetric Redundant ISL Removal (ARIR) algorithm [51] was developed, aiming to eliminate redundant ISLs while preserving overall network capacity. In pursuit of greater autonomy and real-time adaptability, an onboard centralized ISL planning framework for LEO constellations was proposed [52], which improves network stability by effectively reducing end-to-end delays. Addressing the challenge of frequent link failures due to the high mobility of LEO satellites, ReISL [29] introduced a replanning mechanism based on Double DQN, designed to maximize system capacity while minimizing the cost of link reconfiguration. Furthermore, refined LISL planning strategies have been proposed to selectively prune links from the traditional grid-mesh topology, thereby improving average bandwidth without sacrificing overall network availability [44].

3.1.4. Hybrid relay strategies

While ISLs form the backbone of satellite-based networking, several studies have also explored hybrid relay strategies that incorporate ground stations to complement or even substitute ISL connectivity in certain scenarios. For example, one study demonstrated that for low-latency wide-area communication, relying solely on ground-based relays — without any ISL — could still outperform traditional optical fiber networks in terms of latency, particularly over long distances [53]. Further comparative analysis of LEO satellite networks with and without ISL revealed that utilizing ISLs can deliver up to three times higher throughput compared to systems that rely solely on ground relay links [54]. This highlights the performance advantage of incorporating direct satellite-to-satellite communication even when ground relays are available. Additionally, the trade-off between latency and reliability was investigated in LEO mega-constellations by comparing terrestrial, aerial, and space-based relaying strategies. The findings underscore the importance of considering a multi-layered relay architecture, where ground stations and ISLs are jointly optimized to achieve a balance between end-to-end delay, redundancy, and system robustness [55].

In summary, ISL topology design in LEO mega-constellations requires careful consideration of satellite dynamics, visibility constraints, and network performance trade-offs. With the aid of advanced planning, visibility modeling, and hybrid relay strategies, recent research has laid a strong foundation for building scalable and efficient topologies. Building on this, the next section explores ISL scheduling techniques, which determine how and when links should be activated to further optimize network performance under dynamic conditions.

3.2. ISL scheduling

While ISL topology defines the structural framework of satellite connectivity, ISL scheduling determines which ISLs should be activated at any given moment to maintain efficient and reliable communication [56]. In essence, scheduling involves selecting the optimal timing and configuration of active links to adapt to changing network conditions and maximize overall performance.

In LEO satellite networks, where satellites are in constant motion and traffic demands fluctuate rapidly, effective ISL scheduling becomes crucial for ensuring stable connectivity, low latency, and high throughput. To address these challenges, recent research has explored a variety of computational approaches—including metaheuristic algorithms, Integer Linear Programming (ILP) models, and Reinforcement Learning (RL) techniques. Table 2 summarizes the key characteristics, advantages, and limitations of these methodologies. These methods aim to enhance adaptability, optimize resource allocation, and reduce communication delays in highly dynamic satellite environments.

3.2.1. Metaheuristics

Metaheuristic algorithms have gained significant attention in ISL scheduling due to their ability to efficiently explore large and complex solution spaces, providing near-optimal results in dynamic satellite network environments. These methods are particularly well-suited for handling the high variability in topology and traffic demands typical of LEO mega-constellations.

Du et al. [56] proposed a Data-driven Heuristic Assisted Memetic Algorithm (DHMA) for ISL scheduling in the BeiDou Navigation Satellite System (BDS). This approach combines traditional heuristic strategies with evolutionary search mechanisms, enabling faster convergence toward optimal link scheduling. A key strength of DHMA is its adaptive search mechanism, which dynamically adjusts link selection based on real-time traffic conditions, thereby enhancing robustness and ensuring stable scheduling even under variable load patterns.

To further improve link stability in laser-based inter-satellite networks, Bao et al. [57] introduced a Multi-Objective Simulated Annealing (MOSA) method that incorporates refined visibility modeling. Unlike conventional scheduling techniques that often ignore dynamic satellite positions, their approach actively allocates links based on real-time visibility conditions. Additionally, they developed an optimized handover mechanism to reduce disruptions caused by satellite mobility. As a result, their method achieved greater link stability and more balanced traffic distribution, addressing key challenges in dynamic LEO environments.

Liu et al. [58] proposed another metaheuristic approach called Data-Driven Parallel Adaptive Large Neighborhood Search (DP-ALNS), formulated on a time-discrete multi-commodity flow model. In this framework, ISLs are treated as transmission paths and satellites as network nodes, enabling effective link scheduling across large-scale constellations. Their method is particularly suitable for managing the complex traffic flow and time-dependent topology variations inherent in LEO mega-constellations.

3.2.2. Integer Linear Programming (ILP)

In addition to metaheuristic approaches, ILP-based models offer a mathematically rigorous framework for ISL scheduling by explicitly defining both network constraints and performance objectives. These models are particularly well-suited for scenarios that demand precise optimization and predictable performance guarantees.

Wang et al. [59] proposed a Mixed Integer Linear Programming (MILP) method built upon the Time-Expanded Graph (TEG) model, targeting the joint optimization of ISL scheduling and power allocation. The TEG model effectively captures the time-dependent availability of ISLs, enabling accurate representation of dynamic satellite connectivity over discrete time intervals. This formulation supports the efficient allocation of communication resources while minimizing network operational costs.

A notable advantage of this MILP-based approach is its ability to maintain robust and consistent link connectivity, even under the rapidly changing conditions of LEO satellite constellations [60]. By integrating both temporal dynamics and power considerations, the model ensures stable communication performance in highly variable environments.

3.2.3. Reinforcement learning

Recently, RL has emerged as a powerful tool for ISL scheduling, owing to its ability to dynamically adapt link activations based on changing network conditions. Wang et al. [61] introduced an asynchronous hierarchical federated learning framework for dynamic LISL scheduling, specifically tailored for LEO constellations. Unlike traditional centralized methods, their framework allows each satellite to train scheduling models locally, thereby reducing communication overhead and preserving data privacy. A key advantage of this federated learning approach is its support for decentralized optimization, which

Table 2
Approaches for ISL scheduling.

Method	Description	Advantages	Limitations	Sources
Metaheuristics	Heuristic search exploring large solution spaces under dynamic topology.	Near-optimal solutions with fast convergence for large constellations.	No strict optimality guarantees; performance depends on parameter tuning.	[56–58]
Integer Linear Programming (ILP)	Exact optimization using time-expanded graph and linear constraints.	Provides mathematically optimal schedules with clear performance guarantees.	High computational complexity; scalability limited for large constellations.	[59,60]
Reinforcement Learning (RL)	Adaptive policy learning through environment interaction.	Decentralized, real-time decision-making; dynamic adaptability.	Training overhead; potential instability in convergence.	[61–64]

significantly lowers computational complexity while maintaining effective performance. Their study demonstrated improvements in network latency and energy efficiency, reinforcing the feasibility of RL-based distributed scheduling in large-scale satellite networks.

Another work proposed a Multi-Agent Deep Reinforcement Learning (MADRL) framework for the joint optimization of ISL scheduling and routing [62]. This approach enables each satellite to make independent, adaptive link activation decisions in real time, based on observed traffic patterns and network conditions. By supporting decentralized, intelligent decision-making, their method enhances network responsiveness and maintains robust scheduling performance under dynamic traffic loads. The study also highlighted improved resource utilization through continuous learning, showcasing the potential of RL-based methods for integrated LISL scheduling and routing in LEO mega-constellations.

Further advancing this line of research, the study of [63] incorporated network coding into the ISL scheduling framework to enhance multi-hop transmission efficiency. Their method dynamically adjusted link parameters in response to fluctuating traffic demands, reducing data redundancy and optimizing overall resource usage. Simulation results validated the effectiveness of network coding in balancing high reliability with efficient bandwidth utilization, making it a promising enhancement for data transmission in highly dynamic satellite networks.

More recently, research focus has extended to ensuring strict safety guarantees alongside efficiency. For instance, Yi et al. [64] introduced a Safe Reinforcement Learning (Safe RL) approach incorporating a shield mechanism to jointly optimize link scheduling and routing. By enforcing safety constraints, this method achieved zero packet loss due to congestion while improving throughput by 33% and reducing delay by 45% compared to baseline schemes.

In summary, ISL scheduling plays a pivotal role in dynamically managing inter-satellite connectivity under the time-varying conditions of LEO constellations. A wide range of algorithmic approaches — from heuristic optimization to advanced learning-based frameworks — have been proposed to improve link utilization, reduce latency, and enhance network adaptability. Building upon these scheduling strategies, the next section focuses on routing methods that determine how data is forwarded across the dynamically scheduled ISL network to ensure efficient end-to-end communication.

3.3. Routing strategies

Routing in LEO satellite networks presents unique challenges that significantly differ from those in terrestrial environments. These include frequent topology changes, non-negligible propagation delays, and stringent power constraints on satellite hardware. To operate effectively under such conditions, routing protocols must dynamically adapt to real-time link variations, balance traffic loads, and ensure quick recovery from link disruptions or failures.

Conventional *minimum-hop routing* algorithms offer simplicity and ease of implementation but often fail to account for congestion control and link variability—two critical aspects in highly dynamic satellite constellations. To address these limitations, Werner et al. [65] proposed an Asynchronous Transfer Mode (ATM)-based routing approach in LEO/MEO networks, employing a modified Dijkstra's shortest path algorithm that incorporates dynamic link states.

An alternative direction is *traffic-aware, adaptive routing*, exemplified by the Orbit Prediction Shortest Path First (OP-SPF) algorithm [66], which leverages the deterministic nature of orbital motion. OP-SPF further integrates an on-demand mechanism to handle unforeseen link failures. Complementing these efforts, Handley [53] demonstrated that low-latency routing over LISLs can significantly outperform terrestrial optical fiber networks in terms of end-to-end delay—underscoring the importance of ISL-aware routing in next-generation satellite systems.

More recently, *reinforcement learning (RL)-based routing* has gained attention for its ability to learn adaptive forwarding policies through environmental interaction. By considering network states such as link delay or congestion as part of the learning process, RL-based schemes can dynamically optimize routing decisions without explicit modeling of network dynamics. These approaches show promise in improving throughput and robustness under highly dynamic LEO conditions.

In parallel, *on-demand routing* mechanisms aim to ensure reliability and efficiency on unpredictable link failures. These protocols dynamically repair routes as needed, thereby enhancing network connectivity. Such adaptive behavior enables the autonomous and self-healing capabilities essential for next-generation satellite networks.

Additionally, *onboard processing* enables autonomous, real-time routing to reduce ground reliance. To mitigate strict power and weight constraints, lightweight addressing and all-optical bypassing are essential for balancing computational overhead with low-latency demands.

In the following, we explain these routing strategies with ISLs in more detail, and also provide a summary in Table 3.

3.3.1. Minimum-hop routing

Minimum-hop routing has long been a popular approach in satellite networks due to its simplicity, as it prioritizes paths with the fewest intermediate hops to minimize overall end-to-end latency. Ekici et al. [67] introduced a distributed routing algorithm tailored for datagram traffic in LEO satellite networks. Their method dynamically updates routing tables in response to satellite mobility, overcoming the limitations of static routing. By incorporating real-time position awareness, the algorithm minimizes path recalculation delays and demonstrates strong adaptability to the highly dynamic topology of LEO constellations, maintaining consistent performance under frequent configuration changes.

Expanding on this foundation, Wang et al. [78] analyzed ISL characteristics in multi-layer satellite constellations, using a non-homogeneous Poisson distribution to model link variability more accurately. Their study revealed that introducing even a small number of Inter-Layer Links (ILLs) significantly reduces the average hop count across the network. These ILLs enhance routing flexibility and contribute to substantial reductions in end-to-end latency, highlighting their strategic value for routing optimization in large-scale LEO systems.

Parallel to these inter-layer advancements, Elbehiry et al. [79] addressed the connectivity challenges within a single shell, specifically between counter-rotating ascending and descending meshes. While such inter-mesh links are typically neglected due to high relative velocities, they utilized Uniform Cost Search and Bellman-Ford algorithms

Table 3
Routing strategies with ISLs.

Strategy	Mechanism	Advantages	Limitations	Sources
Minimum-hop	Paths with the smallest intermediate hops.	Simple implementation with minimized end-to-end latency.	Ignores network congestion and dynamic link variability.	[67,68]
Traffic-aware and Adaptive	Dynamic path adjustment based on real-time traffic loads.	Improved throughput and QoS satisfaction through load balancing.	Higher computational complexity and control overhead.	[69–71]
RL-based	Routing policy learning from network state and feedback.	Decentralized, autonomous, and adaptive decision-making.	Requires continuous learning updates for dynamic environments.	[72–74]
On-demand	Dynamic path establishment based on real-time conditions.	Control traffic reduction and quick path repair after failures.	Route discovery latency and possible initial packet loss.	[24,75]
Onboard Processing	Performs regenerative processing and routing directly on satellites.	Optimized performance using diverse real-time onboard data.	Introduces significant computational overhead and power consumption.	[76,77]

to schedule feasible connections under strict physical constraints. Their simulation results demonstrated that exploiting these inter-mesh links significantly reduces end-to-end delay and improves resource efficiency compared to standard grid topologies constrained to intra-mesh connections.

Additional theoretical analyses have explored the hop-count distribution in ISL paths across LEO mega-constellations. Chen et al. [68] proposed a model to estimate expected hop counts, providing insights into the structural efficiency of ISL networks. Further work by Chen et al. [80] examined distance characteristics within the minimum-hop path set, comparing shortest-distance and longest-distance paths among those with the same hop count, thereby uncovering trade-offs between hop efficiency and physical path length.

Lee et al. [81] investigated how LISL-based routing is affected by time-varying satellite topology, comparing the routing performance of optical vs. RF-based ISLs. Their findings emphasize the importance of topology-aware routing in achieving reliable performance under real-world satellite motion patterns. Moreover, Chen et al. [82] proposed a gateway placement optimization model using a genetic algorithm to solve an integer optimization problem, aiming to maximize networking performance while satisfying hop-count constraints. Their results demonstrated that strategic gateway placement can significantly enhance overall routing efficiency, especially in networks constrained by latency and coverage requirements.

3.3.2. Traffic-aware and adaptive routing

To further enhance routing performance in LEO satellite networks, numerous studies have explored traffic-aware, optimization-based, and learning-driven adaptive routing strategies.

An explicit load-balancing technique was proposed by Taleb et al. [69], utilizing onboard processing to achieve better traffic distribution in non-GEO satellite networks. Similarly, Mohorcic et al. [70] introduced a traffic class-dependent routing scheme for LEO ISL networks, where routing decisions were made based on three traffic classes — latency-sensitive, throughput-intensive, and best-effort — allowing differentiated handling of diverse service requirements.

Building on optimization-based approaches, Zhao et al. [71] proposed a multi-constraint optimal routing algorithm based on a modified ant colony optimization framework. Their method considers multiple factors such as link bandwidth, transmission delay, node residual storage, and forwarding rate, achieving efficient routing under complex constraints. Huang et al. [83] extended this direction by developing an adaptive multi-path traffic scheduling scheme also based on ant colony optimization, which dynamically selects multiple paths to improve load balancing and routing robustness.

A cooperative routing scheme was developed by Yang et al. [84] to improve data downloading performance via ISLs. Their approach formulated the problem as a maximum-flow-minimum-cost optimization, dynamically allocating feeder link resources under Ka-band bandwidth limitations. This method achieved notable gains in throughput at Earth stations, demonstrating the effectiveness of joint optimization in feeder link-constrained satellite environments.

3.3.3. RL-based routing

RL has emerged as a promising solution for routing in dynamic and large-scale satellite networks, enabling decentralized and adaptive decision-making based on real-time network feedback.

Mao et al. [72] proposed a hierarchical RL-based routing strategy for ultra-dense FSO satellite networks, effectively managing inter-layer traffic and reducing congestion. Their method adaptively distributed data flows, improving routing efficiency and transmission reliability under heavy traffic and frequent topology changes.

Shi et al. [73] introduced a Q-learning-based adaptive routing algorithm designed to support multipath transmission under dynamic traffic conditions. Their approach continuously updated routing policies through learning, enhancing link reliability and network stability despite varying demands.

Zuo et al. [74] developed a DQN-based intelligent routing algorithm tailored for LEO satellite networks. By leveraging deep reinforcement learning, satellites autonomously adjusted routing paths in response to real-time network variations, significantly improving routing responsiveness and real-time decision quality.

In hybrid ISL networks, Zheng et al. [85] proposed a modified Q-routing algorithm designed for laser-microwave heterogeneous links. Their method evaluated real-time link quality to dynamically switch between communication modes, integrating link-quality metrics directly into routing decisions. This adaptive mechanism improved transmission performance and ensured seamless operation across hybrid ISL environments.

3.3.4. On-demand routing

In highly dynamic LEO satellite networks, on-demand routing strategies enhance reliability and adaptability under frequent topology changes. On-demand routing establishes paths dynamically based on current network conditions, enabling responsive data delivery and seamless connectivity. This approach reduces overhead and improves robustness in unpredictable environments.

Stock et al. [75] proposed a distributed on-demand routing scheme specifically designed for the Starlink mega-constellation. Their method efficiently minimized routing overhead by updating paths in real time without relying on centralized precomputation. This dynamic behavior enabled the network to rapidly adapt to topological changes, maintaining consistent performance even in the presence of frequent satellite movement and fluctuating traffic demands.

Bhattacharjee et al. [24] further extended this concept by introducing an on-demand routing framework for dynamic LISLs in LEO mega-constellations. Their approach dynamically adjusted ISL configurations in response to real-time network conditions, significantly reducing the latency associated with establishing new links. A key contribution of their work was the real-time adaptation mechanism, which improved path discovery speed, enhanced throughput, and ensured high availability—all critical features for next-generation satellite networks that demand low-latency and fault-tolerant communication.

To further address routing reliability, Lu et al. [86] proposed a fault-tolerant routing method based on Finite State Automata (FSA),

enabling the network to dynamically reconfigure routes in response to link failures. Additionally, Zeng et al. [87] introduced the Redundant Multi-Path Routing (RMPR) scheme, which enhances fault tolerance and reduces latency by transmitting multiple data copies along shortest edge-disjoint paths. This approach ensures resilient communication, even in the presence of intermittent link disruptions, and is particularly suitable for LEO satellite networks where connectivity conditions change rapidly.

3.3.5. Onboard processing and routing decisions

In large-scale Direct-Satellite-to-Device (DS2D) architectures, lightweight route encoding and advanced onboard processing are essential to keep routing and control overhead manageable despite massive user populations and highly dynamic LEO constellations [76]. To fully exploit the potential of such space-based broadband services and address current latency limitations, regenerative payloads with full onboard protocol-stack processing — including demodulation, decoding, and routing — are increasingly required compared to simple transparent bent-pipe repeaters [76,77]. However, these regenerative architectures involving conventional Decode-and-Forward (DF) relays necessitate full signal recovery and optical-to-electrical-optical (OEO) conversions at each hop, which introduces significant computational overhead and remains fundamentally constrained by the spacecraft's mass, power, and thermal budgets [27].

The routing decisions in these complex systems must leverage diverse information to optimize performance. This includes satellite ephemeris for dynamic topology and handover management, link load status for routing and congestion control, and channel state information (CSI) for precoding, beamforming, and resource management [77]. In constellations with centralized control, routes may be derived from pre-computed snapshots of satellite positions [88], while distributed on-demand schemes can operate with minimal headers that carry only satellite identifiers, leveraging regular mesh-neighbor relations to make purely arithmetic forwarding decisions [75]. Furthermore, advanced techniques such as segment routing via IPv6 extension headers [89], routing-oblivious designs using ground-computed path tags [90], and semantic addressing [91] are employed to shift the computational burden off the satellite and minimize per-hop processing delay.

Beyond traditional networking metrics, routing decisions can also be guided by the physical state of the optical path and the satellite power subsystem. Under translucent optical payload architectures, lightpaths may either transparently bypass relay satellites or undergo OEO regeneration at designated regeneration nodes to maintain an acceptable quality of transmission. In this context, regeneration-routing decisions rely on the Maximum Bypass Hops (MBH), defined from a BER evolution model for repeated LISLs, as well as the current battery depth of discharge of candidate satellites to alleviate aging. The proposed schemes explicitly account for the energy consumed by OEO conversion and digital signal processing at regeneration nodes, and promote optical bypass at intermediate satellites whenever possible in order to avoid unnecessary electronic processing [14].

As an alternative to managing the latency and processing overhead of OEO conversions at relay satellites, all-optical relay schemes aim to keep the signal entirely in the optical domain across intermediate nodes. In particular, Optical Hard Limiter (OHL)–based architectures perform intensity-thresholding directly on OOK signals to suppress noise and restore signal quality without requiring electronic-domain decoding. By eliminating OEO conversions and associated electronic processing at relays, OHL-based all-optical ISL relays target lower end-to-end delay and reduced computational complexity, while their decision thresholds and beam divergence must be adaptively tuned to remain robust under the time-varying link distances and tracking errors of dynamic LEO constellations [27].

In summary, achieving reliable and high-performance networking in LEO satellite systems critically depends on three interconnected pillars: dynamic topology design, efficient ISL scheduling, and robust routing strategies. Recent research has demonstrated that leveraging metaheuristic algorithms, ILP models, and RL techniques can effectively address key challenges arising from the highly dynamic network topology, variable traffic patterns, and limited onboard resources.

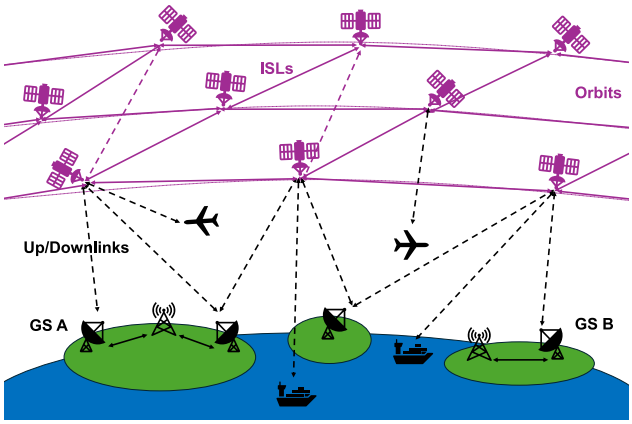


Fig. 7. Satellite networking through ISLs and ground links. The solid and dashed arrows represent persistent links and temporary links, respectively.

These approaches collectively contribute to significant improvements in real-time adaptability, latency reduction, resource efficiency, and network resilience, underscoring their value in the context of large-scale LEO constellations. Looking forward, promising directions for future research include the development of AI-driven predictive scheduling models, hybrid optimization frameworks, and adaptive RL-based routing protocols. Such advances are expected to further enhance the responsiveness, scalability, and fault tolerance of next-generation satellite networks, pushing the boundaries of what is achievable in global space-based communication systems.

4. ISL applications

In this section, we explore practical applications of ISLs in two key satellite system domains: general-purpose satellite communication systems and the Global Navigation Satellite System (GNSS). Fig. 7 shows that ISLs can facilitate global coverage over maritime and aerial regions. However, due to constantly changing network topology, it requires appropriate management techniques for stable connectivity. We highlight recent research efforts that leverage ISL technology as a core enabler to enhance system capabilities, improve performance, and support more advanced and resilient architectures. These case studies illustrate how ISL can play a transformative role in expanding the functionality and efficiency of next-generation satellite infrastructures.

4.1. LEO-based satellite network systems

Several commercial satellite communication services have recently emerged based on LEO mega-constellations. Notable examples include Starlink, Kuiper, OneWeb, and Lightspeed [92]. In contrast to traditional satellite networks that rely on a bent-pipe architecture, most of these modern constellations are capable of forming sophisticated mesh networks through ISLs. This architecture minimizes dependence on ground stations, enabling near-global coverage and low-latency communication through alternative routing paths and the high transmission speed of FSO links compared to terrestrial networks. Several recent constellations exemplify this paradigm; SpaceX's Starlink, for instance, has established a global mesh network in which each satellite is equipped with three ISLs, supporting data rates of up to 200 Gbps [21,46]. With over 7000 LEO satellites in operation, Starlink achieved a 96 Gbps bandwidth capacity [93,94]. Similarly, Amazon's Kuiper constellation which plans to deploy 3236 satellites [95], demonstrated a 100 Gbps

Table 4
LEO satellite network systems and ISL capabilities.

Operator/System	Status	Scale	LISLs per satellite	Data rate per LISL	Architecture
SpaceX/Starlink	Operational	>7,000	3	200 Gbps	Global mesh
Amazon/Kuiper	Planned	3236	N/A	100 Gbps	Global mesh
Telesat/Lightspeed	Planned	198	4	10 Gbps	Global mesh
Eutelsat/OneWeb	Operational	648	None	N/A	Bent-pipe

LISL as part of its mesh network development [96]. Telesat's Lightspeed is designed as a global mesh network with 198 satellites, each featuring four ISLs that provide a 10 Gbps data rate per link [97,98]. Conversely, Eutelsat's OneWeb exemplifies the traditional model, operating a constellation of 648 satellites without ISLs, which rely on ground stations to provide global coverage and connectivity [99]. Table 4 summarizes these representative LEO systems in terms of constellation scale, deployment status, and ISL-enabled architectural features. While these architectural specifications define the theoretical capabilities of LEO constellations, recent empirical studies provide insight into their actual operational performance and the specific behaviors of ISL-enabled networking.

Unlike traditional GSO systems, LEO satellites move at much higher velocities, resulting in a highly dynamic network topology. In such an environment, ISLs plays a crucial role in sustaining high-performance satellite networks by ensuring seamless connectivity and enabling efficient handover mechanisms between fast-moving satellites. Furthermore, ISLs enable LEO mega-constellations to operate as more autonomous space networks, supporting timely inter-satellite data exchange while alleviating the reliance on ground infrastructure [1]. Specifically, ISLs significantly enhance network connectivity. The efficient establishment of inter-plane ISLs maximizes connectivity by accounting for predictable relative motions [9], while inter-layer links bridge different orbital layers to reduce the average hop count [78]. ISLs also accelerate the timeliness of on-board data transmission; satellites can collaboratively offload data to maximize downlink throughput during limited ground contact windows [100], and optimized ISL topologies lower propagation delay by minimizing the network diameter [88]. Moreover, ISLs effectively reduce reliance on ground stations. They allow satellites to route traffic to gateways beyond their immediate line-of-sight, lowering ground deployment costs [82], and enable entirely space-based traffic forwarding to bypass geographically or politically constrained regions [54]. Finally, ISLs improve the capability for inter-satellite status information exchange. This allows adjacent satellites to collaboratively learn spatiotemporal traffic correlations for resilient on-board traffic prediction [101] and supports near-continuous networked telemetry monitoring that overcomes the visibility limitations of traditional ground-based architectures [102].

4.1.1. System status and performance analysis

The rapid deployment of LEO mega-constellations has transitioned from theoretical design to operational reality, necessitating empirical performance analysis. The Starlink architecture leverages an extensive ground segment, featuring more than 200 ground stations and 31 Points of Presence (PoPs) to support its LEO operations [103]. The network services over 2 million subscribers across 63 countries, utilizing a massive fleet of satellites and aiming to provide global coverage through a large LEO constellation [104]. Starlink's second-generation (Gen2) satellites fully adopt LISLs, enabling onboard signal processing and space-based routing and thereby reducing reliance on dense ground-station deployments [105].

Recent measurement studies provide granular insights into the performance of these networks under various conditions. Crowdsourced measurements show that Starlink typically achieves median download goodput between 100 and 200 Mbps with median round-trip times of about 40–50 ms [106]. However, performance varies with mobility and environmental blockages; for instance, a recent measurement campaign with the Flat High Performance terminal in Northern Europe reported high-speed, low-latency connectivity, while showing modest reductions in median throughput and slightly increased RTT during in-motion scenarios compared to stationary use, mainly due to occasional outages caused by roadside obstructions [107]. Highway measurements showed that mobile Starlink experiences reduced throughput and increased packet loss compared to stationary use, and that short total obstructions from highway bridges cause pronounced burst losses and RTT spikes [108]. Similarly, in-flight performance evaluations report median downlink and uplink throughputs of 64 Mbps and 24 Mbps, respectively, though significant uplink degradation was observed during aircraft descent below 17,000 feet [109]. Environmental conditions also impact link quality; the WetLinks dataset, collected over a span of six months, reveals a statistically significant but weak negative correlation between precipitation intensity and download throughput [110]. To predict these variations, a learning-based system called StarNet is proposed, which forecasts user throughput in Starlink by modeling satellite mobility, traffic load dynamics, and weather conditions using multivariate time-series prediction [111].

The integration of ISLs allows Starlink's bent-pipe architecture to extend over multiple satellites, improving connectivity for remote regions while introducing distinctive routing challenges. Measurements on Réunion Island show that ISL-enabled routing yields round-trip latencies of around 150 ms, outperforming the roughly 210 ms observed with incumbent terrestrial services [112]. Similarly, empirical data from Hawaii showed an average latency of 77 ms via ISLs, surpassing the 108 ms latency of terrestrial submarine cables [103]. However, this extended bent-pipe model, which uses in-space relays via ISLs, still exhibits periodic behavior: globally synchronized network reconfigurations roughly every 15 s induce brief sub-second variations in latency and throughput that are visible at fine time scales [104]. Moreover, users who rely exclusively on these links — such as maritime clients — often encounter RTTs that are significantly higher than the theoretical minimum due to extended routing paths [113]. In some edge cases, such as for users in the Indian Ocean region, Starlink's current ISL routing, which always sends traffic back to the user's home PoP before heading toward the destination, can lead to RTTs exceeding 500 ms, highlighting the impact of such indirect routing paths [114].

Comparative analyses highlight the trade-offs between different constellation architectures. Unlike Starlink, the OneWeb network, which in its current Phase 1 deployment does not utilize ISL capabilities, generally fulfills its throughput Service Level Agreements (SLAs) but exhibits notable latency variations during Satellite Network Portal handovers in certain geographical regions [115]. Despite this, field trials in Northern Finland showed that OneWeb's latency and jitter are sufficiently stable to support mission-critical public safety applications, including real-time video services [116]. Recent measurements comparing terrestrial 5G with Starlink and OneWeb LEO networks indicate that, while terrestrial 5G generally outperforms individual satellite links, integrating multiple links through multi-connectivity (e.g., 5G–satellite and satellite–satellite) can markedly reduce outage probabilities in the studied scenarios compared to using any single link alone [117]. For live streaming over LEO satellite networks, a middleware called SARA has been proposed, which mitigates rebuffering caused by frequent satellite handover–induced outages by dynamically adjusting playback speed and assisting existing ABR algorithms [118]. Emerging services like Starlink's Direct-to-Cell are also being characterized; while the current beta service is SMS-only, initial physical-layer measurements suggest an approximate per-beam

downlink capacity of 3.1 Mb/s as an upper-bound average throughput, with observed signal strengths noticeably lower than those of terrestrial mobile networks [105].

4.1.2. ISL-assisted data downlinking

To address the aforementioned bottlenecks and improve data throughput, several studies have explored collaborative data downloading techniques that leverage ISLs. In [100], a method was proposed to enhance downlink performance by enabling satellites to offload data to neighboring satellites via ISLs prior to establishing contact with ground stations. This approach allows the network to optimize the utilization of ground contact windows, thereby significantly increasing overall data downloading throughput. Simulation results confirmed the effectiveness of this technique in maximizing downlink efficiency under limited ground access conditions.

In addition to ISL-based collaboration, spectrum-level optimizations have also been investigated. A downlink spectrum sharing scheme based on the Upper Confidence Bound (UCB) index was introduced in [119–121], aiming to maximize throughput while satisfying interference constraints. Building on this, Yun et al. [122] proposed a Dynamic Downlink Interference Management method using Deep Q-Network (DQN) reinforcement learning. Their approach adapts to time-varying interference environments without requiring direct inter-satellite communication, and demonstrated performance close to the theoretical optimum.

4.1.3. Handover strategies

Due to the high orbital velocity of LEO satellites, it is inherently difficult to maintain long-duration connections between satellites and ground stations. As a result, handover mechanisms play a critical role in ensuring seamless communication and maintaining network reliability within satellite network systems.

To address this, Tong et al. [123] proposed a Satellite Handover scheme considering the Link Outage Probability (SHLOP) within a Satellite-UAV Integrated Network (SUIN). This approach takes into account the intermittent failures of both ISLs and Satellite-to-Ground Links (SGLs). By modeling the probability of link outage, SHLOP determines the optimal allocation of reserved resources during handover, thereby maximizing handover success rates while minimizing unnecessary resource consumption.

In a related study, Salem et al. [124] investigated uplink soft handover techniques for LEO constellations. Assuming that both the serving and target satellites can receive uplink signals from ground users during handover, their results showed that soft handover achieved a lower block error rate compared to traditional hard handover, highlighting its potential for improving transmission reliability during satellite transitions.

Additionally, Yang et al. [23] proposed a grouped LISL handover scheme along with a routing method called Open Shortest Path First-Predictive Update Routing (OSPF-PUR). This approach models the dynamic satellite topology as a Finite State Automaton (FSA), enabling more proactive and stable handover management across changing network states.

4.1.4. Resource allocation and energy efficiency

Resource allocation poses a significant challenge in satellite communication systems due to the limited availability of power and frequency resources, particularly for establishing and maintaining active ISLs. While LISLs offer advantages such as lower interference susceptibility and higher energy efficiency compared to RF-based ISLs, both technologies face constraints in deployment under dynamic network conditions [1].

To address these limitations, Zeng et al. [102] proposed a resource allocation method for networked telemetry systems utilizing ISLs between MEO and LEO satellites to monitor LEO mega-constellations. Their approach simultaneously tackled access scheduling and subchannel–power allocation by applying a block coordinate descent optimization method, improving resource utilization across network layers.

Focusing on the power-latency trade-off in FSO satellite networks, Liang et al. [125] introduced a Binary Integer Linear Programming (BILP) model to systematically optimize power allocation. Their method enabled satellites to minimize transmission energy while meeting stringent latency constraints, striking an effective balance across intercontinental optical links. Simulation results highlighted that optimization-driven resource management is essential for energy-constrained mega-constellations, achieving efficient performance without compromising transmission speed.

To further address variable service requirements and topology dynamics, Tong et al. [126] proposed the Dynamic Even distribution with Network Coding (DENC) method for resource scheduling in LEO satellite networks. DENC accounts for intermittent ISL availability and ensures efficient resource usage by distributing traffic loads evenly and applying network coding to enhance reliability and throughput.

4.1.5. Ultra-low latency applications

Another promising application of LISLs lies in supporting ultra-low latency communications for time-critical services. Singh et al. [127] investigated the feasibility of using LISL-based satellite constellations for high-frequency trading (HFT). Their study demonstrated that such networks could connect major global financial hubs and, in many cases, outperform conventional terrestrial fiber networks in terms of latency. This highlights the potential of LISLs to enable high-speed global financial transactions, where even milliseconds of latency reduction can offer significant advantages. However, the study also emphasized that in-satellite processing delays can partially negate the benefits of low-latency LISL transmission. This finding underscores the importance of optimizing onboard processing pipelines in tandem with link-layer performance.

Beyond financial networks, ISLs have also been considered for global air traffic control and in-flight passenger communications. Hermenier et al. [128] developed a delay model to assess the impact of various routing policies on end-to-end latency in airborne communication scenarios. Their work provides a foundational understanding of how ISL-based routing can influence service responsiveness in safety-critical and commercial aviation contexts.

In summary, the integration of ISLs into LEO satellite communication systems has demonstrated significant advantages across various dimensions, including data throughput, handover efficiency, energy-aware resource allocation, and ultra-low latency communication. These advancements highlight ISL's critical role in overcoming the unique challenges of large-scale, dynamic satellite networks. In the following section, we explore how ISL technologies are also being leveraged in Global Navigation Satellite Systems (GNSS) to enhance positioning accuracy, system robustness, and global coverage.

4.2. Global navigation satellite system (GNSS)

The Global Navigation Satellite System (GNSS) is a critical space-based infrastructure that enables precise positioning, navigation, and timing (PNT) services on a global scale. GNSS supports a wide range of applications, including aviation, maritime navigation, military operations, and civilian services. Multiple independent GNSS constellations are currently in operation, including the Global Positioning System (GPS) maintained by the United States, BeiDou Navigation Satellite System (BDS) by China, GLONASS by Russia, and Galileo by the European Union.

The GNSS constellations exhibit varied adoption of ISLs. The GPS constellation, with 31 operational MEO satellites, utilizes RF crosslinks to enhance system operations [129,130]. BDS employs ISLs in its hybrid constellation of 24 MEO, 3 GEO, and 3 inclined GSO satellites, enabling communication across different orbital layers [131]. GLONASS operates

Table 5
ISL configurations in GNSS.

System	ISL Implementation	ISL Status	No. Satellites
GPS	RF Crosslinks	Active	31
BDS	RF Inter-layer	Active	30
GLONASS	RF/Laser Hybrid	Active	26
Galileo	None	Planned	27

a constellation of 26 satellites, evolving its ISL capabilities from S-band RF links on the 21 M-series and pulsed laser systems on the 4 K-series to the latest K2, which integrates both technologies alongside high-precision atomic clocks to enhance navigation performance [132,133]. In contrast, the Galileo system, consisting of 27 MEO satellites, does not currently feature ISL capabilities but plans to introduce them in its second-generation constellation [132,134]. We summarize the ISL configurations of these GNSS constellations in Table 5.

4.2.1. Precise orbit determination using ISL

One of the most critical technologies determining the performance of GNSS is Precise Orbit Determination (POD), which enables satellites to calculate their orbital positions with high accuracy. Accurate POD is essential for improving the positioning precision of GNSS services, especially for high-precision applications such as geodesy, autonomous navigation, and scientific measurement.

Traditionally, POD has relied on ground station networks to collect satellite observation data and compute orbital parameters. However, this approach has limitations due to geographical constraints, uneven ground station distribution, and delayed updates. To reduce dependence on ground infrastructure, recent research has focused on autonomous navigation techniques that allow satellites to determine their orbits independently. In this context, ISLs have emerged as a powerful enabler, facilitating time and position synchronization between satellites within a constellation and enabling onboard POD computation.

A notable example is China's BeiDou-3 (BDS-3) system, where ISLs are used to enhance autonomous orbit determination. Studies have shown that the integration of ISLs improved 3D orbit accuracy by more than 50% compared to traditional ground-based methods [135]. Moreover, the use of ISLs ensured that orbit accuracy remained stable even when satellites moved beyond the visibility of ground stations [136].

Similar research has been conducted on the Galileo constellation, where simulation-based studies examined the impact of ISLs on orbit and clock estimation performance [137]. By varying ISL coupling schemes and sampling intervals, the researchers demonstrated that ISL integration reduced orbit determination errors and enabled stable performance with a smaller number of ground stations.

Beyond satellite-centric improvements, ISLs also enhance Precise Point Positioning (PPP) accuracy for ground users. In the BDS-3 constellation, the addition of ISLs reduced ground-based positioning errors, as ISL-enabled POD led to more accurate orbit and clock information for correction [138].

ISLs have also been explored in the context of broadband LEO constellations for navigation [139]. Due to the higher satellite density and faster orbital periods in LEO, these systems present new opportunities for high-frequency POD updates. A simulation study demonstrated that LEO constellations, when enhanced with ISLs, can achieve more precise orbit determination than MEO or HEO constellations, particularly when integrated with globally distributed ground station networks [134].

4.2.2. ISL-based topology optimization for GNSS

Beyond orbit estimation accuracy, several studies have explored ISL topology optimization and link assignment strategies to further improve positioning precision and communication efficiency in GNSS.

Liu et al. [140] proposed a link assignment framework for a laser/radio hybrid GNSS system, aiming to improve Position Dilution of Precision (PDOP) by leveraging the complementary characteristics of LISL and RF-based ISLs. Building on this, Zeng et al. [141] introduced a multi-objective discrete binary particle swarm optimization approach for ISL topology optimization in GNSS, with the goal of minimizing PDOP, link delay, and terminal idle rate.

Several works have applied genetic algorithms (GAs) to optimize ISL configurations. Huang et al. [142] designed a GA-based optimization scheme to improve orbit determination performance and reduce delay by adjusting time slice lengths and polling cycles. Similarly, Chengzhuo et al. [143] presented a multi-objective GA framework for dynamic LISL topology management in GNSS, balancing communication performance and orbit estimation accuracy by minimizing both link transmission cost and PDOP.

Further, Yan et al. [144] proposed a joint topology optimization scheme for time-slotted ISLs and Ground-Satellite Links (GSLs) under a Polling Time Division Duplex (PTDD) mechanism. Their model formulates the problem as an Integer Linear Programming (ILP) problem and introduces an efficient heuristic algorithm to minimize data delivery latency while preserving orbit determination accuracy.

4.2.3. ISL support for autonomous GNSS navigation

Looking toward the next generation of GNSS, autonomous navigation has emerged as a key objective to reduce dependence on ground infrastructure and enhance positioning resilience. ISLs have been identified as a core enabler of such functionality through their ability to support distributed coordination, precise ranging, and time synchronization.

To this end, Yan et al. [145] proposed a distributed Contact Plan Design (CPD) framework for autonomous navigation, which was successfully simulated on the BeiDou constellation. Their approach allows satellites to coordinate communication schedules independently, enabling more efficient orbit determination and reduced reliance on centralized ground control.

In a complementary direction, Luo et al. [146] explored the potential of adding lunar satellites to the GNSS framework to support extended orbit determination capabilities. Their findings suggest that incorporating satellites in lunar orbit can improve GNSS autonomy and POD accuracy, especially in deep space or coverage-challenged regions.

Further, Sun et al. [147] developed link assignment algorithms for time synchronization and ranging using ISLs. Their simulations on BeiDou-3 showed that the proposed algorithm can achieve optimal ISL utilization within limited time slots, while simultaneously enhancing orbit estimation accuracy, which is crucial for onboard autonomous navigation.

4.2.4. Conclusion of ISL applications in satellite systems and GNSS

Given the inherent limitations of time, frequency, and power resources in space environments, efficient resource allocation mechanisms, including link scheduling and power-aware optimization, are essential for the sustainable operation of satellite network systems. In particular, future research must carefully assess the relative utility of ISLs, taking into account the cost and availability of frequency bands and onboard energy constraints.

ISL has proven to be a key enabler of greater autonomy, enhanced throughput, and reduced reliance on ground stations in LEO-based satellite communications. Looking forward, further advancements are expected in areas such as efficient power management, autonomous networking protocols, and the convergence of Non-Terrestrial Networks (NTN) with terrestrial infrastructure to support a wide range of emerging application services.

At the same time, the importance of ISL in enhancing GNSS performance continues to grow. From POD to autonomous navigation and time synchronization, ISL enables GNSS constellations to become more accurate, robust, and self-reliant. As GNSS evolves toward more distributed and intelligent architectures, optimization-driven ISL designs will remain a vital area of research for improving both performance and global availability.

5. Challenges and future directions

5.1. Practical challenges in LISL

Practical deployment of LISLs faces significant hurdles starting at the optical physical layer. Harsh space environments — exposure to solar radiation, plasma, and extreme temperature variations — affect optical alignment and component reliability, while the narrow beamwidth of lasers makes ISLs highly sensitive to satellite platform vibrations and pointing errors [3]. Specifically, vibrations during attitude adjustments or from internal mechanisms can severely degrade link stability or even cause interruptions [148]. These LISL signals are mainly impaired by pointing errors due to satellite jitter and tracking noise, which introduces a trade-off between concentrating the beam energy and mitigating misalignment loss [149]. Consequently, maintaining precise line-of-sight in the presence of satellite vibration, accurately handling point-ahead angles, and compensating for the large Doppler dynamics in high-speed space links remain persistent difficulties for optical inter-satellite communication systems [3].

At the communication and link control level, highly precise PAT mechanisms are required to establish and maintain LISLs between satellites in crossing orbital planes, where high relative velocities and stringent SWaP (Size, Weight, and Power) constraints make stable connectivity particularly challenging [13]. The most critical bottleneck is the PAT delay — currently ranging from several to tens of seconds — which renders highly dynamic link switching unfeasible for delay-sensitive applications and problematic for temporary link formation in large constellations [15]. These limitations are further compounded by restricted payload and power budgets that allow only a finite number of optical transceivers per satellite, while such lengthy PAT procedures severely hinder the practical deployment of highly dynamic LISLs in current systems [150]. Consequently, fully enabling transient LISLs will require next-generation optical systems capable of achieving millisecond-scale PAT latency [15].

At the network and system level, LISL design must balance low end-to-end latency against stringent power and optical path-loss constraints, because tighter transmission-power limits reduce the maximum feasible LISL range and thereby restrict low-latency routing options [125]. Furthermore, instability in link establishment and outages caused by satellite dynamics, vibration, and pointing errors complicate LISL topology design and routing efficiency in mega-constellation networks, motivating research on acquisition-probability-based probabilistic link models and topology optimization for outage-prone links [150]. Due to these constraints, current mega-constellations are often limited to semi-static, grid-like topologies, and inter-plane ISLs tend to be less utilized because of their frequent handovers and link interruptions [1].

Finally, at the application and implementation level, future optical ISL terminals are expected to adopt advanced, compact beam-steering technologies — such as Optical Phased Arrays (OPAs) and Micro-Electro-Mechanical Systems (MEMS) scanners — to gradually replace bulky gimbal-based mechanisms and better satisfy stringent SWaP constraints [16]. These emerging technologies aim to overcome the limitations of traditional mechanical optics while improving beamwidth adjustment accuracy. Furthermore, the integration of tunable liquid lenses enables real-time beam divergence control with response times of 1–10 ms, allowing rapid compensation for dynamic link distance variations without mechanical movement [27]. Beyond compact beam-steering and terminal design, future optical ISL systems must also optimize routing and scheduling under rapidly changing LEO topologies, while further strengthening security against eavesdropping and exploring the integration of quantum key distribution and other quantum-secure schemes [16].

5.2. Future research directions

The extreme complexity and high dynamics of LEO mega-constellations render traditional routing protocols increasingly inefficient [72]. Consequently, recent research has shifted toward Artificial Intelligence (AI) and Machine Learning (ML) techniques to more effectively manage network topology, ISLs, and traffic in future space networks [92]. Graph Neural Networks (GNN) and Deep Reinforcement Learning (DRL) have been jointly leveraged to capture time-varying topological characteristics and to generate adaptive inter-satellite routing decisions that reduce end-to-end delay and improve load balancing [151]. Notably, Multi-Agent DRL (MADRL) enables decentralized decision-making at the satellite level, allowing individual satellites to dynamically schedule LISLs and balance traffic loads [62,152,153]. Furthermore, a safe RL framework has been developed that enforces safety constraints in link scheduling and routing to effectively avoid packet loss and improve reliability for mission-critical services [64].

The integration of LEO satellites into 3GPP Non-Terrestrial Network (NTN) standards toward 6G is driving the adoption of DS2D communications [76,154]. Specifically, emerging Direct-to-Cell technologies leverage advanced onboard processing to support unmodified smartphones, necessitating more sophisticated space-based network management [105]. In this architecture, ISLs serve as a pivotal space-based backhaul, connecting user terminals in remote areas to terrestrial core networks. Moving beyond conventional bent-pipe architectures, NTN-based satellite networks are evolving toward regenerative payloads, enabling ISLs to carry inter-satellite interfaces (e.g., Xn interface). This evolution facilitates seamless handovers and cooperative resource management directly in orbit. As DS2D services transition from basic messaging to broadband data, robust ISL meshes will be essential to handle aggregated traffic loads and ensure ubiquitous coverage independent of ground station proximity [76].

Early research on LEO satellite networks primarily relied on simulation tools, such as Hypatia [12], to model ISL topology and routing. However, recent studies highlight a paradigm shift toward empirical measurements using operational mega-constellations like Starlink. Novel methodologies, such as HitchHiking [113], enable large-scale active measurement of LEO satellite network latency and routing characteristics while bypassing the need for specialized satellite hardware or user recruitment. Consequently, future research must leverage large-scale, longitudinal datasets (e.g., WetLinks [110]) to analyze the impact of environmental factors and orbital dynamics on LEO satellite network performance and reliability, thereby bridging the gap between simulation and reality.

6. Conclusion

This survey has provided a comprehensive overview of ISL technologies, with a focus on their communication, networking, and application aspects. We began by examining the fundamental characteristics of ISL communications, including RF and laser technologies, and explored the technical challenges related to signal processing and link establishment. We then reviewed recent advancements in ISL networking, particularly in scheduling strategies and routing frameworks, highlighting the importance of dynamic topology management and connectivity maintenance. Finally, we discussed emerging applications and the integration of ISL-enabled architectures into future satellite internet systems.

This work has concentrated on the technical dimensions of ISL design and operation, while broader issues such as regulation, safety, and security in large-scale ISL deployment remain beyond its scope. Throughout this paper, we have emphasized that effective ISL design is essential for achieving high-performance satellite networks, especially in the context of LEO mega-constellations. As satellite systems continue to expand in scale and complexity, ISLs will play a central role in enhancing reliability, reducing latency, and minimizing

reliance on ground infrastructure. Furthermore, we identified practical hurdles in LISL deployment, such as PAT latency and SWaP constraints, and outlined the roadmap toward AI-driven management and 6G integration.

Looking ahead, future research should address open challenges across ISL communication, networking, and application domains. In the communication domain, improving link reliability and adapting modulation schemes to dynamic environments remain key objectives. Networking research should advance real-time topology management and intelligent scheduling, while application-level studies should focus on seamless integration with terrestrial and non-terrestrial systems. As global demand for low-latency, high-throughput connectivity grows, coordinated progress across all ISL aspects will be vital to enabling next-generation space-based networks.

CRediT authorship contribution statement

Byoungkyu Ji: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Junghyun Han:** Writing – original draft, Investigation. **Daeun Kim:** Writing – original draft, Investigation. **Jihyeon Yun:** Writing – review & editing. **Changhee Joo:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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